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Exercises on Binary Numbers and Boolean Logic

Introduction to Computer Programming

Management and Artificial Intelligence

LUISS

Recap: Binary Numbers

Number Systems: Positional vs. Non-Positional

- **Non-Positional Systems**: The value of a symbol is independent of its position.
    - Example: In the Roman numeral `XXVIII` (28)
        - both `X` symbols represent 10

- **Positional Systems**: The value of a symbol depends on its position within the number.
    - Example: In the decimal number `37579`:
        - the first `7` represents 7000
        - while the second `7` represents 70

General Positional System

- Any natural number $b > 1$ can be used as the **base** of a positional number system.
- A number is written as a sequence of digits $d_{n-1}d_{n-2}...d_1d_0$, where each digit d_i is an integer from 0 to $b-1$.
- The value of the number is calculated using the formula:

$$N = \sum_{i=0}^{n-1} d_i \times b^i = d_{n-1}b^{n-1} + d_{n-2}b^{n-2} + \cdots + d_1b^1 + d_0b^0$$

The Binary System (Base 2)

When the base $b = 2$, we get the **binary system**. The only possible digits are 0 and 1.

Example: Convert the binary number 11011_2 to decimal

$$(11011)_2 = 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$$

$$= 16 + 8 + 0 + 2 + 1 = (27)_{10}$$

Bits and Bytes

- **Bit**: A single binary digit (0 or 1).
- **Byte**: A group of 8 bits.

With one byte, we can represent $2^8 = 256$ different values (from 0 to 255). This is enough to represent all the characters on a standard keyboard using an encoding like ASCII.

Octal and Hexadecimal Systems

These systems are commonly used in computing as a compact way to represent binary numbers.

- **Octal**: Uses $b = 8$ and digits 0-7.
- **Hexadecimal**: Uses $b = 16$ and digits 0-9 and letters A-F, where:
  - `A = 10`, `B = 11`, `C = 12`, `D = 13`, `E = 14`, `F = 15`

Converting from Decimal to Another Base

To convert a decimal number N_{10} to base b, we use the method of **successive divisions**:

1. Divide N_{10} by the base b. The remainder is the rightmost digit.
2. Replace N_{10} with the quotient from the division.
3. Repeat until the quotient is 0. The remainders, read in reverse order, form the number in base b.

Example: Convert $\mathbf{116_{10}}$ to Binary

$116 \div 2 = 58 \Rightarrow$ remainder **0**

$58 \div 2 = 29 \Rightarrow$ remainder **0**

$29 \div 2 = 14 \Rightarrow$ remainder **1**

$14 \div 2 = 7 \Rightarrow$ remainder **0**

$7 \div 2 = 3 \Rightarrow$ remainder **1**

$3 \div 2 = 1 \Rightarrow$ remainder **1**

$1 \div 2 = 0 \Rightarrow$ remainder **1**

Reading the remainders from bottom to top, we get: $(1110100)_2$.

Example: Convert 1547_{10} to Hexadecimal

$1547 \div 16 = 96 \Rightarrow$ remainder **11** (which is `B` in hex)

$96 \div 16 = 6 \Rightarrow$ remainder **0**

$6 \div 16 = 0 \Rightarrow$ remainder **6**

Reading the remainders from bottom to top, we get: $(60B)_{16}$.

Exercises

Possible exam questions!

Decimal 24 to Binary

Convert the decimal number **24** to binary:

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$24 \div 2 = 12$ (rem 0)

Decimal 24 to Binary

Convert the decimal number **24** to binary:

$24 \div 2 = 12$ (rem 0)

$12 \div 2 = 6$ (rem 0)

Decimal 24 to Binary

Convert the decimal number **24** to binary:

$24 \div 2 = 12$ (rem 0)

$12 \div 2 = 6$ (rem 0)

$6 \div 2 = 3$ (rem 0)

Decimal 24 to Binary

Convert the decimal number **24** to binary:

$24 \div 2 = 12$ (rem 0)

$12 \div 2 = 6$ (rem 0)

$6 \div 2 = 3$ (rem 0)

$3 \div 2 = 1$ (rem 1)

Decimal 24 to Binary

Convert the decimal number **24** to binary:

$24 \div 2 = 12$ (rem 0)

$12 \div 2 = 6$ (rem 0)

$6 \div 2 = 3$ (rem 0)

$3 \div 2 = 1$ (rem 1)

$1 \div 2 = 0$ (rem 1)

Decimal 24 to Binary

Convert the decimal number **24** to binary:

$24 \div 2 = 12$ (rem 0)

$12 \div 2 = 6$ (rem 0)

$6 \div 2 = 3$ (rem 0)

$3 \div 2 = 1$ (rem 1)

$1 \div 2 = 0$ (rem 1)

Result: $(11000)_2$

Decimal 63 to Binary

To convert 63 from decimal to binary:

Decimal 63 to Binary

To convert 63 from decimal to binary:

$63 \div 2 = 31$ remainder 1

$31 \div 2 = 15$ remainder 1

$15 \div 2 = 7$ remainder 1

$7 \div 2 = 3$ remainder 1

$3 \div 2 = 1$ remainder 1

$1 \div 2 = 0$ remainder 1

Reading the remainders from bottom to top: 111111

Therefore, $63_{10} = 111111_2$

Decimal 129 to Binary

To convert 129 from decimal to binary:

Decimal 129 to Binary

To convert 129 from decimal to binary:

$129 \div 2 = 64$ remainder 1

$64 \div 2 = 32$ remainder 0

$32 \div 2 = 16$ remainder 0

$16 \div 2 = 8$ remainder 0

$8 \div 2 = 4$ remainder 0

$4 \div 2 = 2$ remainder 0

$2 \div 2 = 1$ remainder 0

$1 \div 2 = 0$ remainder 1

Reading the remainders from bottom to top: 10000001

Therefore, $129_{10} = 10000001_2$

Binary 10101 to Decimal

Convert the binary number **10101** to decimal:

Binary 10101 to Decimal

Convert the binary number **10101** to decimal:

$$\begin{aligned}(10101)_2 &= 1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 \\ &= 16 + 0 + 4 + 0 + 1 = (21)_{10}\end{aligned}$$

Binary 100100 to Decimal

Convert the binary number **100100** to decimal:

Binary 100100 to Decimal

Convert the binary number **100100** to decimal:

$$(100100)_2 = 1 \cdot 2^5 + 0 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 0 \cdot 2^0$$

$$= 32 + 0 + 0 + 4 + 0 + 0 = (36)_{10}$$

Binary 11100111 to Decimal

Convert the binary number **11100111** to decimal:

Binary 11100111 to Decimal

Convert the binary number **11100111** to decimal:

$$(11100111)_2 = 1 \cdot 2^7 + 1 \cdot 2^6 + 1 \cdot 2^5 + 0 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$$

$$= 128 + 64 + 32 + 0 + 0 + 4 + 2 + 1 = (231)_{10}$$

Hex 1547 to Decimal

Convert the hexadecimal number **1547** to decimal:

Hex 1547 to Decimal

Convert the hexadecimal number **1547** to decimal:

$$(1547)_{16} = 1 \cdot 16^3 + 5 \cdot 16^2 + 4 \cdot 16^1 + 7 \cdot 16^0$$

$$= 1 \cdot 4096 + 5 \cdot 256 + 4 \cdot 16 + 7 \cdot 1$$

$$= 4096 + 1280 + 64 + 7 = (5447)_{10}$$

Hex CAFE to Decimal

Convert the hexadecimal number **CAFE** to decimal:

Hex CAFE to Decimal

Convert the hexadecimal number **CAFE** to decimal:

$$(CAFE)_{16} = C \cdot 16^3 + A \cdot 16^2 + F \cdot 16^1 + E \cdot 16^0$$

Hex CAFE to Decimal

Convert the hexadecimal number **CAFE** to decimal:

$$(CAFE)_{16} = C \cdot 16^3 + A \cdot 16^2 + F \cdot 16^1 + E \cdot 16^0$$

$$= 12 \cdot 4096 + 10 \cdot 256 + 15 \cdot 16 + 14 \cdot 1$$

Hex CAFE to Decimal

Convert the hexadecimal number **CAFE** to decimal:

$$(CAFE)_{16} = C \cdot 16^3 + A \cdot 16^2 + F \cdot 16^1 + E \cdot 16^0$$

$$= 12 \cdot 4096 + 10 \cdot 256 + 15 \cdot 16 + 14 \cdot 1$$

$$= 49152 + 2560 + 240 + 14 = (51966)_{10}$$

Hex DEADBEEF to Decimal

Convert the hexadecimal number **DEADBEEF** to decimal:

$$DEADBEEF_{16} = D \cdot 16^7 + E \cdot 16^6 + A \cdot 16^5 + D \cdot 16^4 + B \cdot 16^3 + E \cdot 16^2 + E \cdot 16^1 + F$$

$$= 13 \cdot 16^7 + 14 \cdot 16^6 + 10 \cdot 16^5 + 13 \cdot 16^4 + 11 \cdot 16^3 + 14 \cdot 16^2 + 14 \cdot 16^1 + 15 \cdot 16^0$$

$$= 13 \cdot 268435456 + 14 \cdot 16777216 + 10 \cdot 1048576 + 13 \cdot 65536 + 11 \cdot 4096 + 14 \cdot 256 + 14$$

$$= 3489660928 + 234881024 + 10485760 + 851968 + 45056 + 3584 + 224 + 15 = 3735928$$

Recap: Boolean Logic

Boolean Operators and Truth Tables

Boolean algebra operates on truth values (True/1, False/0) using logical operators.

- **AND (Conjunction, $\wedge$)**: True only if both operands are true.
- **OR (Disjunction, $\vee$)**: True if at least one operand is true.
- **NOT (Negation, $\neg$)**: Inverts the truth value.

Truth Table Example

Let's evaluate the expression: $F = (A \land B) \lor (\neg A \land C)$

Truth Table Example

Let's evaluate the expression: $F = (A \land B) \lor (\neg A \land C)$

A	B	C	A AND B	!A	!A AND C	F
0	0	0	0	1	0	**0**
0	0	1	0	1	1	**1**
0	1	0	0	1	0	**0**
0	1	1	0	1	1	**1**
1	0	0	0	0	0	**0**
1	0	1	0	0	0	**0**
1	1	0	1	0	0	**1**
1	1	1	1	0	0	**1**

Key Concepts in Boolean Logic

- **Equivalence**: Two expressions are equivalent if they have the same truth table.
  - *Example (De Morgan's Law)*: $\neg(A \lor B) \equiv \neg A \land \neg B$
- **Tautology**: An expression that is always true, regardless of the input values.
  - *Example*: $A \lor \neg A$
- **Contradiction**: An expression that is always false.
  - *Example*: $A \land \neg A$

Exercises

Formula $(\neg A \land \neg B) \lor (\neg B \land A)$

Formula $(\neg A \land \neg B) \lor (\neg B \land A)$

A	B	!A	!B	!A AND !B	!B AND A	(!A AND !B) OR (!B AND A)
0	0	1	1	1	0	1
0	1	1	0	0	0	0
1	0	0	1	0	1	1
1	1	0	0	0	0	0

Formula $(\neg A \wedge B \wedge \neg C) \vee (A \wedge \neg B)$

Formula $(\neg A \land B \land \neg C) \lor (A \land \neg B)$

A	B	C	!A	!B	!C	!A AND B	!A AND B AND !C	A AND !B	(!A AND B AND !C) OR (A AND !B)
0	0	0	1	1	1	0	0	0	0
0	0	1	1	1	0	0	0	0	0
0	1	0	1	0	1	1	1	0	1
0	1	1	1	0	0	1	0	0	0
1	0	0	0	1	1	0	0	1	1
1	0	1	0	1	0	0	0	1	1
1	1	0	0	0	1	0	0	0	0
1	1	1	0	0	0	0	0	0	0

Formula $A \wedge \neg B \wedge (C \vee B \wedge \neg C) \vee \neg C$

Formula $A \wedge \neg B \wedge (C \vee B \wedge \neg C) \vee \neg C$

A	B	C	!B	!C	B AND !C	C OR (B AND !C)	A AND !B AND (C OR (B AND !C)) OR !C
0	0	0	1	1	0	0	1
0	0	1	1	0	0	1	0
0	1	0	0	1	1	1	1
0	1	1	0	0	0	1	0
1	0	0	1	1	0	0	1
1	0	1	1	0	0	1	1
1	1	0	0	1	1	1	1
1	1	1	0	0	0	1	0

Formula

$$(\neg A \land ((B \land A) \lor (\neg B \land \neg A))) \lor (\neg B \land (A \lor (\neg A \land B)))$$

Formula

$$(\neg A \land ((B \land A) \lor (\neg B \land \neg A))) \lor (\neg B \land (A \lor (\neg A \land B)))$$

A	B	!A	!B	!A AND ((B AND A) OR (!B AND !A))	!B AND (A OR (!A AND B))	(!A AND ((B AND A) OR (!B AND !A))) OR (!B AND (A OR (!A AND B)))
0	0	1	1	1	1	1
0	1	1	0	0	0	0
1	0	0	1	0	1	1
1	1	0	0	0	0	0

Is formula $A \vee \neg(A \wedge B)$ a tautology?

Is formula $A \lor \neg(A \land B)$ a tautology?

A	B	A AND B	NOT(A AND B)	A OR NOT(A AND B)
0	0	0	1	1
0	1	0	1	1
1	0	0	1	1
1	1	1	0	1

Is formula $A \vee \neg(A \wedge B)$ a tautology?

A	B	A AND B	NOT(A AND B)	A OR NOT(A AND B)
0	0	0	1	1
0	1	0	1	1
1	0	0	1	1
1	1	1	0	1

Yes, it is a tautology!

Is formula $A \vee (\neg A \wedge B)$ a contradiction?

Is formula $A \vee (\neg A \wedge B)$ a contradiction?

A	B	!A	!A AND B	A OR (!A AND B)
0	0	1	0	0
0	1	1	1	1
1	0	0	0	1
1	1	0	0	1

Is formula $A \vee (\neg A \wedge B)$ a contradiction?

A	B	!A	!A AND B	A OR (!A AND B)
0	0	1	0	0
0	1	1	1	1
1	0	0	0	1
1	1	0	0	1

No, it is not a contradiction!